Anubhav Gupta

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Education

Thapar Institute of Engineering & Technology — Patiala, India
Bachelor of Engineering in Computer Science; ***CGPA: 8.78 / 10*** — *Aug 2023 – July 2027*

Experience

Machine Learning Research Intern — June 2025 – Aug 2025
Experimental Learning Activity Centre, TIET — *Patiala, Punjab*

- Collaborated in a 4-member research team to develop an end-to-end Python pipeline for processing and classifying multi-channel EEG signals using signal-processing and ML techniques.
- Engineered an end-to-end data pipeline processing multi-channel EEG datasets at scale, extracting 760 signal features across 19 channels.
- Applied K-NN based feature selection to reduce dimensionality to 28 optimal features, improving model efficiency and inference speed.
- Trained and evaluated ML/DL models, including CNN+LSTM with VMD, achieving 93.1% accuracy (AUC: 0.97), and contributed to a comparative research study on ML/DL methods for mental-health analysis.

Projects

NEURIS – Cognitive Memory System | *FastAPI, Neo4j, Redis, Docker, LLMs, Vector Search* — GitHub

- Built an end-to-end cognitive memory platform with a Python backend and React frontend, enabling context-aware retrieval in under 2 seconds using graph relationships and RAG.
- Developed a FastAPI + SQLAlchemy backend, integrating Google Cloud Speech for diarization and Gemini for structured memory extraction.
- Designed Neo4j-based graph storage with Redis caching and Docker Compose orchestration, supporting 3–5 concurrent speakers with consistent identity tracking.
- Modeled episodic memories through timeline clustering, improving retrieval relevance by 35% while maintaining efficient semantic search.

Calivio – Event Management Platform | *Next.js, TypeScript, MongoDB, Stripe* — GitHub | Live

- Built a full-stack SaaS platform for event discovery, hosting, and ticketing, developing backend workflows for users, events, and orders.
- Implemented Stripe Checkout and webhook-driven order processing for reliable payment and transaction handling.
- Designed MongoDB-backed APIs with search, filtering, and server-side pagination to efficiently handle large event datasets.
- Integrated Clerk authentication and developed responsive interfaces using Tailwind CSS and shadcn/UI.

Flight Management System | *MySQL, PL/SQL* — 2025

- Developed a relational database system for airline operations, including ticket booking, passenger management, and flight scheduling.
- Optimized SQL queries to reduce response time by 35% under concurrent-user workloads.
- Implemented stored procedures and triggers to automate ticket pricing based on demand and seasonal trends.
- Designed database workflows to maintain data consistency and reliable transaction processing across multiple operations.

Technical Skills

Languages: C/C++, Python, JavaScript, TypeScript, SQL
Backend & Frameworks: Node.js, FastAPI, Next.js, React.js, REST APIs, NumPy, Pandas, Scikit-learn
Databases: MySQL, MongoDB, Neo4j, Redis
Cloud & Developer Tools: Linux, Git/GitHub, Postman, VS Code, AWS, Docker
Coursework: Operating Systems, Computer Networks, OOP, DBMS, Data Structures, AI & ML
Problem Solving: Solved 600+ DSA problems across different platforms (Codolio Profile)

Achievements & Miscellaneous

- Advanced to Round 2 of Google Big Code Challenge 2026, placing among Top 1,500 students nationwide.
- Ranked among Top 100 teams in ZS Campus Beats 2026.
- AIR 7410 – Goldman Sachs India Hackathon 2026.